WEAKLY ASSOCIATED POINTS OF A FINITE FLAT PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Weakly associated primes under a flat pullback. Divisors, label divisors-lemma-weakly-ass-pullback, invoked at proof line 325. Gives the easy implication: a weakly associated prime downstairs pulls back to one upstairs because the morphism is flat.
- A finite flat algebra over a local ring is free, and weakly associated points under a finite morphism. Algebra, label algebra-lemma-finite-flat-local (proof line 336); Divisors, label divisors-lemma-weakly-associated-finite (proof line 348). The first gives that the localised algebra is a free module of some rank n at least one, so that the pushforward of the pulled back module has the same weakly associated primes as the original downstairs, and the second pushes a weakly associated point through the finite morphism.

Source reference. Original lemma 00NZ: sources/stacks-chow/tags.txt, source lines 877–877. Navigation pointer for \ref{algebra-lemma-finite-flat-local} used inside the counted span (line 336); the argument that consumes it is retained in the span.

Source reference. Original lemma 05F0: sources/stacks-chow/tags.txt, source lines 7009–7009. Navigation pointer for \ref{divisors-lemma-weakly-ass-pullback} used inside the counted span (line 325); the argument that consumes it is retained in the span.

Source reference. Original lemma 05EZ: sources/stacks-chow/tags.txt, source lines 7008–7008. Navigation pointer for \ref{divisors-lemma-weakly-associated-finite} used inside the counted span (line 348); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $g : T \to S$ be a finite flat morphism of schemes. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_S$-module. Let $t \in T$ be a point with image $s \in S$. Then*

$$t \in \mathit{WeakAss}(g^*\mathcal{G}) \Leftrightarrow s \in \mathit{WeakAss}(\mathcal{G})$$

Proof. The implication "$\Leftarrow$" follows immediately from Divisors, Lemma 05F0. Assume $t \in \mathrm{WeakAss}(g^*\mathcal{G})$. Let $\mathrm{Spec}(A) \subset S$ be an affine open neighbourhood of s. Let $\mathcal{G}$ be the quasi-coherent sheaf associated to the A-module M. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to s. As g is finite flat we have $g^{-1}(\mathrm{Spec}(A)) = \mathrm{Spec}(B)$ for some finite flat A-algebra B. Note that $g^*\mathcal{G}$ is the quasi-coherent $\mathcal{O}_{\mathrm{Spec}(B)}$-module associated to the B-module $M \otimes_A B$ and $g_*g^*\mathcal{G}$ is the quasi-coherent $\mathcal{O}_{\mathrm{Spec}(A)}$-module associated to the A-module $M \otimes_A B$. By Algebra, Lemma 00NZ we have $B_\mathfrak{p} \cong A_\mathfrak{p}^{\oplus n}$ for some integer $n \geq 0$. Note that $n \geq 1$ as we assumed there exists at least one point of T lying over s. Hence we see by looking at stalks that

$$s \in \mathrm{WeakAss}(\mathcal{G}) \Leftrightarrow s \in \mathrm{WeakAss}(g_*g^*\mathcal{G})$$

Now the assumption that $t \in \mathrm{WeakAss}(g^*\mathcal{G})$ implies that $s \in \mathrm{WeakAss}(g_*g^*\mathcal{G})$ by Divisors, Lemma 05EZ and hence by the above $s \in \mathrm{WeakAss}(\mathcal{G})$. □